Series & Sequences

Cauchy Criterion

$\{s_n\}$ converges $\Leftrightarrow$ for all $\varepsilon > 0$, $\exists N$ s.t. $m, n \geq N \Rightarrow |s_m - s_n| < \varepsilon$.

For series: $|s_m - s_n| = \left|\sum_{k=n+1}^{m} a_k\right|$. Use this when you can't find the limit explicitly.

Cor. *$\sum a_n$ converges $\Rightarrow a_n \to 0$. (Converse fails: harmonic series.)*

Monotone Sequences

Thm. *Monotone + bounded $\Rightarrow$ convergent.*

To find the limit L: take $n \to \infty$ in the recurrence and solve for L using continuity of the right-hand side.

Key Limits (all $\to 0$)

Proof technique: **Bernoulli's inequality**: $(1+x)^n \geq 1 + nx$ for $x > -1$. Use it to sandwich the sequence between 0 and something that $\to 0$.

$\frac{1}{n^p}$	$p > 0$
$\sqrt[n]{p} \to 1$	$p > 0$: set $x_n = \sqrt[n]{p} - 1$; Bernoulli gives $p \geq (1+x_n)^n \geq 1 + nx_n$
$\sqrt[n]{n} \to 1$	set $x_n = \sqrt[n]{n} - 1 \geq 0$; $n = (1+x_n)^n \geq \binom{n}{2} x_n^2$, so $x_n \leq \sqrt{2/(n-1)}$
$\frac{n^\alpha}{(1+p)^n}$	$\alpha \in \mathbb{R}$, $p > 0$: exponential beats poly
x^n, $\lvert x\rvert < 1$	geometric decay

Convergence Tests

Thm (Comparison). *$|a_n| \leq c_n$ and $\sum c_n$ converges $\Rightarrow \sum a_n$ converges.*
$a_n \geq d_n \geq 0$ and $\sum d_n$ diverges $\Rightarrow \sum a_n$ diverges.

Thm (Cauchy Condensation). *$a_n \geq a_{n+1} \geq 0$: $\sum a_n$ converges $\Leftrightarrow \sum 2^k a_{2^k}$ converges.*

Use for p-series: $\sum 2^k (2^k)^{-s} = \sum 2^{k(1-s)}$, geometric iff $s > 1$.

Thm (Root Test). $\alpha = \limsup |a_n|^{1/n}$. *$\alpha < 1 \Rightarrow$ converges. $\alpha > 1 \Rightarrow$ diverges. $\alpha = 1$: inconclusive.*

Thm (Ratio Test).

$$\beta = \limsup \left|\frac{a_{n+1}}{a_n}\right|, \quad \alpha_* = \liminf \left|\frac{a_{n+1}}{a_n}\right|.$$

$\beta < 1 \Rightarrow$ converges. $\alpha_* > 1 \Rightarrow$ diverges. Use when factorials/exponentials appear.

Key fact: Root test is strictly stronger than ratio test. If ratio test gives L, root test also gives L. Ratio can fail ($L = 1$) where root succeeds.

Power Series

Def. *$\sum c_n (x-a)^n$ has **radius of convergence***

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}.$$

Thm (Cauchy–Hadamard). *Converges absolutely for $|x-a| < R$, diverges for $|x-a| > R$. Check endpoints separately.*

Why $\limsup$*?* Root test: $\limsup |c_n(x-a)^n|^{1/n} = |x-a| \limsup |c_n|^{1/n} < 1 \Leftrightarrow |x-a| < R$.
Examples: $e^x = \sum x^n/n!$: $R = \infty$. $\sum n!\, x^n$: $R = 0$.

Differentiation

Definition & Basic Rules

$$f'(x) = \lim_{t\to x} \frac{f(t) - f(x)}{t - x}.$$

Differentiable $\Rightarrow$ continuous ($C^1 \subset C^0$).
Chain rule: $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$.

Simple Discontinuities

Def. *f has a **simple discontinuity** at x if $f(x^+)$ and $f(x^-)$ both exist but: (i) $f(x^+) \neq f(x^-)$ (jump), or (ii) $f(x^+) = f(x^-)$ but $\neq f(x)$ (removable). Otherwise: discontinuity of the **second kind** (e.g., $\sin(1/x)$).*

Mean Value Theorems

Thm (MVT). *f cts on $[a,b]$, diff'able on (a,b): $\exists c$ with $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.*

Thm (Generalized MVT). *$\exists c \in (a,b)$: $[f(b)-f(a)]g'(c) = [g(b)-g(a)]f'(c)$.*

Proof: Apply MVT to $h = [f(b)-f(a)]g - [g(b)-g(a)]f$, which has $h(a) = h(b)$.

Thm (L'Hôpital). *$f, g \to 0$ as $x \to a$ and $\lim f'/g' = A \Rightarrow \lim f/g = A$.*

Proof: GMVT gives $f(x)/g(x) = f'(c)/g'(c)$ for some $c \in (a,x)$.

Thm (IVT for Derivatives). *If $f'(a) < \lambda < f'(b)$, $\exists x \in (a,b)$ with $f'(x) = \lambda$.*
Proof: Apply EVT to $h(t) = f(t) - \lambda t$; interior min has $h' = 0$.

Taylor's Theorem

$$f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(\beta-\alpha)^k + \frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$$

for some x between α and β (Lagrange remainder).

Proof: Define M by $f(\beta) = P(\beta) + M(\beta-\alpha)^n$. Apply MVT n times to $g(t) = f(t) - P(t) - M(t-\alpha)^n$, using $g^{(k)}(\alpha) = 0$ at each step.

Fixed Points

Thm. *$f'(t) \neq 1$ everywhere $\Rightarrow$ at most one fixed point.*
Proof: If $f(x) = x, f(y) = y$, MVT gives $f'(c) = 1$, contradiction.

Thm (Contraction Mapping). *$|f'(t)| \leq A < 1$ everywhere $\Rightarrow$ unique fixed point $x = \lim x_n$ (any x_1).*

Riemann Integration

Darboux Sums

Partition $P = \{x_0, \ldots, x_n\}$. $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$.

$$U(P,f) = \sum M_i \Delta x_i, \quad L(P,f) = \sum m_i \Delta x_i.$$

$f \in \mathscr{R} \Leftrightarrow \inf_P U(P,f) = \sup_P L(P,f)$.

Thm (Integrability Criterion). *$f \in \mathscr{R} \Leftrightarrow$ for all $\varepsilon > 0$, $\exists P$ with $U(P,f) - L(P,f) < \varepsilon$.*

Thm (Zero Sets). *E is a **zero set** if $\forall \varepsilon > 0$, $\exists$ countable intervals covering E with total length $< \varepsilon$.*
f bounded + disc'ties form a zero set $\Rightarrow f \in \mathscr{R}$.

Key examples: f monotone $\Rightarrow f \in \mathscr{R}$. f continuous $\Rightarrow f \in \mathscr{R}$. Dirichlet function $\notin \mathscr{R}$.

Riemann–Stieltjes

Replace Δx_i by $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$.

Thm. *$\alpha' \in \mathscr{R}$, f bounded: $f \in \mathscr{R}(\alpha) \Leftrightarrow f\alpha' \in \mathscr{R}$ and*

$$\int f\, d\alpha = \int f\alpha'\, dx.$$

Proof: MVT gives $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$; compare Stieltjes sum to Riemann sum for $f\alpha'$, error $\leq M[U(P,\alpha') - L(P,\alpha')]$.

Thm (Integration by Parts). $\int_a^b f\, dg + \int_a^b g\, df = f(b)g(b) - f(a)g(a)$.

Thm (Change of Variable). *$\varphi : [A,B] \to [a,b]$ strictly increasing, $\beta = \alpha \circ \varphi$, $g = f \circ \varphi$:*

$$\int_A^B g\, d\beta = \int_a^b f\, d\alpha.$$

Improper Integrals & Hölder

Def. *$\int_a^\infty f\, dx = \lim_{R\to\infty} \int_a^R f\, dx$ (if limit exists).*

Thm (Comparison). *$0 \leq f \leq g$: $\int g$ converges $\Rightarrow \int f$ converges.*

Key: $\int_1^\infty x^{-p}\, dx$ converges iff $p > 1$. $\int_0^1 x^{-p}\, dx$ converges iff $p < 1$.

Thm (Hölder's Inequality). *$\frac{1}{p} + \frac{1}{q} = 1$, $p, q > 1$:*

$$\int_a^b |fg|\, dx \leq \left(\int_a^b |f|^p\right)^{1/p} \left(\int_a^b |g|^q\right)^{1/q}.$$

Proof: Young's inequality $uv \leq u^p/p + v^q/q$ (from concavity of ln). Normalize $u = |f|/\|f\|_p$, $v = |g|/\|g\|_q$, integrate.
$p = q = 2$: Cauchy–Schwarz. *To apply:* split $h = f \cdot g$ cleverly, choose p, q so both $\int |f|^p$ and $\int |g|^q$ are finite.

Distance & Metric Continuity

$\rho_E(x) = \inf_{e\in E} d(x,e)$. Properties:

- $\rho_E(x) = 0 \Leftrightarrow x \in \bar{E}$ (uses: $x \in \bar{E}$ iff every ball hits E).
- $|\rho_E(x) - \rho_E(y)| \leq d(x,y)$ (Lipschitz-1, hence uniformly cts).

Thm (Urysohn-type). *A, B disjoint closed: $f(p) = \rho_A(p)/(\rho_A(p) + \rho_B(p))$ is continuous, $f|_A = 0$, $f|_B = 1$.*
Key: denom > 0 since $p \in A \cap B = \emptyset$ is impossible.

KEY TECHNIQUES

Technique: AM-GM for iteration. To show a Newton-type sequence $x_{n+1} = \frac{1}{p}\big((p-1)x_n + \alpha/x_n^{p-1}\big)$ is bounded below by $\alpha^{1/p}$: view x_{n+1} as the arithmetic mean of $(p-1)$ copies of x_n and one copy of α/x_n^{p-1}. AM-GM gives $x_{n+1} \geq (x_n^{p-1} \cdot \alpha/x_n^{p-1})^{1/p} = \alpha^{1/p}$. Then monotone decrease follows from $x_n - x_{n+1} = (x_n^p - \alpha)/(px_n^{p-1}) > 0$.

Technique: Monotone + bounded → limit. Once you know $\{x_n\}$ is monotone and bounded, it converges. To find the limit L: take $n \to \infty$ in the recurrence. Since f is continuous, $L = f(L)$. Solve for L.

Technique: MVT as a bounding tool. The MVT gives $|f(b) - f(a)| = |f'(c)||b-a| \leq \sup|f'| \cdot |b-a|$. Use this whenever you need to bound a function value in terms of its derivative. Especially powerful when $f(a) = 0$: then $|f(x)| \leq M_1(x-a)$.

Technique: Bootstrapping / Gronwall. You have $M_0 \leq c \cdot M_0$ where $c < 1$ (e.g., $c = A(x_0 - a) < 1$). Conclude $M_0 = 0$. Then repeat on the next interval with the new base point. This "bootstraps" a local result ($f \equiv 0$ near a) into a global result ($f \equiv 0$ on $[a,b]$) by covering the interval in finitely many steps of length $\leq 1/(2A)$.

Pattern: $f(a) = 0$, $|f'| \leq A|f| \Rightarrow |f(x)| \leq AM_0(x_0 - a) \cdot (M_0/M_0) \Rightarrow M_0 \leq A(x_0 - a)M_0 \Rightarrow M_0 = 0$.

Technique: Auxiliary function / uniqueness by subtraction. To prove uniqueness (or that something is zero): suppose f and g both satisfy the conditions. Define $h = f - g$. Then h satisfies $h(a) = 0$ and $|h'| \leq A|h|$ (by Lipschitz). Apply Gronwall to conclude $h \equiv 0$.

General pattern: reduce "$f = g$" to "$h = f - g$ satisfies conditions implying $h \equiv 0$".

Technique: Contraction mapping (fixed points). $|f'| \leq A < 1$ everywhere $\Rightarrow$ iterate $x_{n+1} = f(x_n)$. Then $|x_{n+1} - x_n| \leq A^{n-1}|x_2 - x_1|$. Tail sum: $|x_m - x_n| \leq \frac{A^{n-1}}{1-A}|x_2 - x_1| \to 0$, so $\{x_n\}$ is Cauchy. Limit satisfies $f(x) = x$ by continuity.

Technique: Triangle inequality for ρ_E. To show $|\rho_E(x) - \rho_E(y)| \leq d(x,y)$: for any $e \in E$, use triangle inequality $d(x,e) \leq d(x,y) + d(y,e)$ then take inf over e. This gives $\rho_E(x) \leq d(x,y) + \rho_E(y)$. Swap x, y for the other direction. Abstract principle: taking inf over a third variable preserves triangle inequality.

Technique: Creating functions from metric (Urysohn). To continuously separate two disjoint closed sets A, B: use $f = \rho_A/(\rho_A + \rho_B)$. The denominator is nonzero since $p \in A \cap B$ is impossible (they're disjoint closed). Then f is a ratio of continuous functions with positive denominator, so f is continuous. Properties of ρ_E from part (a) then immediately give $f|_A = 0$ and $f|_B = 1$.

Technique: Abel summation (sums ↔ integrals). To show $\sum 1/n^s = s\int_1^\infty [x]/x^{s+1}\,dx$: write $s\int_n^{n+1} n/x^{s+1}\,dx = n(n^{-s} - (n+1)^{-s})$. Sum and telescope. The telescoping gives $\sum_{n=1}^N n^{-s} - N^{1-s}$; let $N \to \infty$.

General: Abel summation = "summation by parts": $\sum a_n(b_n - b_{n+1}) = a_N b_N - a_1 b_1 + \sum(a_{n+1} - a_n)b_{n+1}$.

Technique: Eps-delta recipe. (1) Write out what you need: $d(f(p), f(q)) < \varepsilon$. (2) Work backwards: express $d(f(p), f(q))$ in terms of $d(p,q)$ using triangle inequality and available bounds. (3) Read off δ: if $d(p,q) < \delta$ implies $d(f(p), f(q)) \leq C \cdot \delta$, set $\delta = \varepsilon/C$.

For metric spaces: to show f is cts at p, find δ s.t. $d(p,q) < \delta \Rightarrow d(f(p), f(q)) < \varepsilon$. For uniform cts, δ must be independent of p.

Technique: Partition refinement (integrability). To show $f \in \mathscr{R}$: construct a partition P with $U(P,f) - L(P,f) < \varepsilon$. Strategy: isolate the "bad" subintervals (where f varies a lot) into a set of small total length. On the rest, f varies by at most ε.

Formal: bad subintervals contribute $\leq 2M \cdot$ (total bad length); good ones contribute $\leq \varepsilon \cdot (b-a)$.

Technique: Comparison for improper integrals. $\int_1^\infty |f(x)|/x^{s+1}\,dx$: if $|f(x)| \leq C$ (bounded), then the integral is $\leq C\int_1^\infty x^{-(s+1)}\,dx = C/s$, which converges for $s > 0$. This handles the $x - [x]$ term in the zeta function proof.

Technique: Triangle inequality in L^2 (square → Schwarz → root). To prove $\|f - h\|_2 \leq \|f-g\|_2 + \|g-h\|_2$: (1) set $u = f-g$, $v = g-h$, so $f - h = u + v$; (2) square both sides — reduces to showing $\|u+v\|_2^2 \leq (\|u\|_2 + \|v\|_2)^2$; (3) expand left side and right side — everything cancels except the cross term:

$$\int uv\,d\alpha \;\leq\; \|u\|_2\|v\|_2;$$

(4) apply Schwarz ($p = q = 2$ Hölder) to bound the cross term; (5) take square root.

Template: square, expand, bound cross term by Schwarz, factor as perfect square, root.

Technique: L^2 approximation by piecewise linear interpolation (Ex. 12). To construct continuous g with $\|f - g\|_2 < \varepsilon$ for $f \in \mathscr{R}(\alpha)$: (1) since $f \in \mathscr{R}(\alpha)$, pick P with $U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon^2/(2M)$; (2) define g by linear interpolation between the values of f at partition points — g is continuous since adjacent pieces agree at shared endpoints; (3) on each $[x_{i-1}, x_i]$, both $f(t)$ and $g(t)$ (a convex combination of f-values) lie in $[m_i, M_i]$, so $|f - g| \leq M_i - m_i$; (4) integrate:

$$\begin{aligned}\|f-g\|_2^2 &\leq \sum (M_i - m_i)^2\,\Delta\alpha_i \\ &\leq 2M \underbrace{\sum (M_i - m_i)\Delta\alpha_i}_{=U-L} \\ &< \varepsilon^2.\end{aligned}$$

Key trick: $(M_i - m_i)^2 \leq 2M(M_i - m_i)$ — one factor of oscillation is absorbed into the global bound $2M$, converting squared oscillations to linear ones so the Riemann criterion applies.

Quick Reference

Convergence Summary

$\sum 1/n^s$	conv. iff $s > 1$
$\sum r^n$	conv. iff $\lvert r\rvert < 1$
$\sum 1/(n\log n)$	diverges
$\int_1^\infty x^{-p}$	conv. iff $p > 1$
$\int_0^1 x^{-p}$	conv. iff $p < 1$

Zeta Function Facts

For $s > 1$:

$$\zeta(s) = s\int_1^\infty \frac{[x]}{x^{s+1}}\,dx = \frac{s}{s-1} - s\int_1^\infty \frac{x - [x]}{x^{s+1}}\,dx.$$

Second integral converges for all $s > 0$ by comparison with $\int x^{-(s+1)}$.

Key Inequalities

- AM-GM: $\frac{1}{p}\sum a_i \geq (\prod a_i)^{1/p}$
- MVT: $|f(b) - f(a)| \leq \sup|f'| \cdot |b - a|$
- Lipschitz: $|f(x) - f(y)| \leq A|x - y|$
- Young: $uv \leq u^p/p + v^q/q$ $(1/p + 1/q = 1)$
- Hölder: $\int |fg| \leq \|f\|_p\|g\|_q$
- Cauchy–Schwarz: $|\langle u, v\rangle| \leq \|u\|\|v\|$

MVT Applications

- $f(a) = f(b) \Rightarrow f'(c) = 0$ (Rolle's)
- $f' = 0$ on $(a,b) \Rightarrow f$ constant
- $f' \geq 0$ on $(a,b) \Rightarrow f$ nondecreasing
- $|f'| \leq A|f|$ and $f(a) = 0 \Rightarrow f \equiv 0$ (Gronwall)
- Lipschitz ODE $\Rightarrow$ unique solution (via Gronwall)